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# Minimizing appearance of Global Positioning System drift locations in Emergency vehicle Tracking System using haversine distance

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***Abstract*—Assigning, monitoring vehicles in real time becomes important in an Emergency Vehicle Management System. The visual Geographic Information System mapping from the Global Positioning System data gives the idea of the current location of the vehicle. There is always a possibility of showing wrong location of the vehicle due to Global Positioning System drift. Due to this drift at times stationary vehicle may be shown in the map as moving. The drift may be because of various environmental factors and signal interference. The requirement to minimize this drift is a necessity. A novel approach in which the locations are filtered out by calculating the haversine distance between location data of two successive packets is presented here in this paper. Different thresholds are considered for stationary and running vehicle for filtering. The resultant data after filtration is used for plotting. This results in minimum drifted locations being plotted on the map.**

***Keywords—Global Positioning System, Geographic Information System, Emergency Vehicle Management System, haversine distance, location coordinates, map***

## I. Introduction

Vehicle Tracking system for Emergency vehicles uses Geographic Information System (GIS) and also data from Smart Phone, Global Positioning System (GPS) and Mobile Data Terminal (MDT). Geographic Information System mapping allows users to visualize the location of a vehicle on a geographic map, to assign vehicles for dispatch and also monitor the vehicle movement in real time. A GIS map is used in this application to show the location coordinates of vehicle movement. This helps to dispatch a vehicle quickly and also assures low response time and better performance. However the drift occurring in Global Positioning System (GPS) gives inaccurate live location of the object. A drift is the deviation or error in position of vehicle. This GPS drift occurs in several cases: when the vehicle is stationary, moving with low speed, environmental conditions, signal interference or going under cover. The location can be incorrectly positioned ahead, behind, left or right due to drift. When this data is plotted on a map and displayed for the user, the visual gives the wrong location coordinates instead of actual location coordinates. This misinformation may mislead the user and also results in delay in vehicle selection. So a method for filtering that would remove unwanted or drifted locations becomes a necessity before plotting the location coordinates data on the map. The unwanted locations are filtered by using haversine distance between two successive location data in packets with respective to Coordinated Universal Time (UTC) . When the distance is less than certain threshold the locations are marked invalid and are not plotted. The threshold for vehicle in running state and halt state are different. By this way the drifted locations are removed to a larger extent even though hundred percent removal was not achieved.

Haversine distance is employed in the calculation of the nearest or closest distance between two location coordinates. The novelty of the paper lies in using it effectively for removing GPS drifted locations.

## II. Literature Survey

### A. Related Work

The haversine formula is utilized for the calculation of distance between locations by several authors. For several applications like finding the best or optimal route between two locations, distance analysis for marketing, recommendations, reservations etc.

V. Hegde, T. S. Aswathi and R. Sidharth focussed on location wise distance analysis [1]. They have used haversine formula to calculate distance and k-means for clustering. The method aimed at analysing student residential distance for admission analysis.

S. R. Senthilkumar, V. H. Jabak, G. Vidyashankar, D. Divij and S. Saravanan have used haversine distance to find the best Bus route aiming at the Educational Institutes [2].

Y. Dian Harja and R. Sarno found the nearest medical services using the haversine distance [3]. The setup is using Haversine algorithm to discover medical services surrounding it within a definite radius.

K. Saputra, N. Nazaruddin, D. H. Yunardi and R. Andriyani [4] implemented haversine formula for their Location Based Mobile Application. The solution provided is an Android-based application. This system gives entire details regarding the location of buildings.

R. R. Safitri, A. Pratiarso and A. Zainudin came up with a system that is also based on the Haversine formula. The application uses it to compute the distance from the current location of the vehicle to the parking area [5]. This smart parking reservation system is also a mobile based application.

F. A. Hizham and R. V. H. Ginardi used the Haversine Formula to choose a clinic within a particular distance [6]. The system decides the nearby clinic recommendations by using the information retrieval method and the Haversine Formula.

S. Al Farisi, Fauziah and R. Tamara Aldisa applied the Haversine Formula in their application. In addition they used

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sequential searching for their web based online attendance system [7]. Two algorithms are used here to reduce absenteeism and cheating.

M. A. Talha et al. used haversine formula for in case of emergencies [8]. A key innovation is the employment of the Haversine formula to determine the nearest police station, enabling faster response times during emergencies.

### *B. Haversine Formula*

The great circle distance is the shortest distance between two points on the surface of a sphere measure along the arc. But it has considerable error for very small distances. So the accurate distance between two points on the spherical surface of the earth is calculated by the haversine formula. The haversine formula uses the latitude, longitude to find the shortest distance between two points. The trigonometric function haversine can be expressed in terms of cosine function [9].

The haversine function (1) is given as:

$$haversine(\theta)=sin^2(\theta/2) \quad (1)$$

where $\theta$ is the angle

The haversine distance measures the angular distance between two points .

"The simplified Haversine formula (2) when latitude and longitude are given in radians:

$$d=2r \arcsin \{ \sqrt{(\sin^2[(\varphi_2-\varphi_1)/2]+\cos(\varphi_1).\cos(\varphi_2)(\sin^2[(\lambda_2-\lambda_1)/2]))}\} \quad (2)$$

with:

d=the distance between two points (along the large circle of the sphere)

r=radius of the earth, 6371 km

$\Phi_1,\varphi_2$=latitude 1 and latitude 2, in radians

$\lambda_1,\lambda_2$=longitude of point 1 and longitude of point 2, in radians

Application of Haversine Formula (3) in the IT field can be made in the form of the following Haversine algorithm.

$$d=2r \arcsin\{ \sqrt{(\sin^2[(\varphi_2-\varphi_1)/2]+\cos(\varphi_1).\cos(\varphi_2)(\sin^2[(\lambda_2-\lambda_1)/2]))}\} \quad (3)$$

The haversine formula should be broken down into the following variables:

$X=\sin^2[(\varphi_2-\varphi_1)/2]$

$Y=\sin^2[\lambda_2-\lambda_1)/2]$

$K=X+\cos(\varphi_1).\cos(\varphi_2)+Y$

$L=\arcsin[\text{sqrt }(K)]$

Convert degree decimal to radius:

$M=\text{asin}(L)$

The result:

d=2r.L (already in meters)" [9].

The disadvantage of haversine function is the relative error of up to 0.5% in calculation. For one kilometre distance the error is approximately five metres. This error is too large for precision applications. But for applications like maps this error is acceptable and hence haversine formula can be used in calculation for drift locations filtering.

## III. Approach And Method Followed

The data should be cleaned before proceeding to filtering. The following method was used. Fig. 1 in detail shows the steps involved in the whole process of filtering out invalid location coordinates for plotting on the map.

The location information becomes a necessity and is required nowadays in many computing or mobile applications [10]. The location coordinates are obtained here from GPS, smart phone and Mobile Data Terminal. All the location coordinates derived from the three devices are subjected to the same process before plotting.

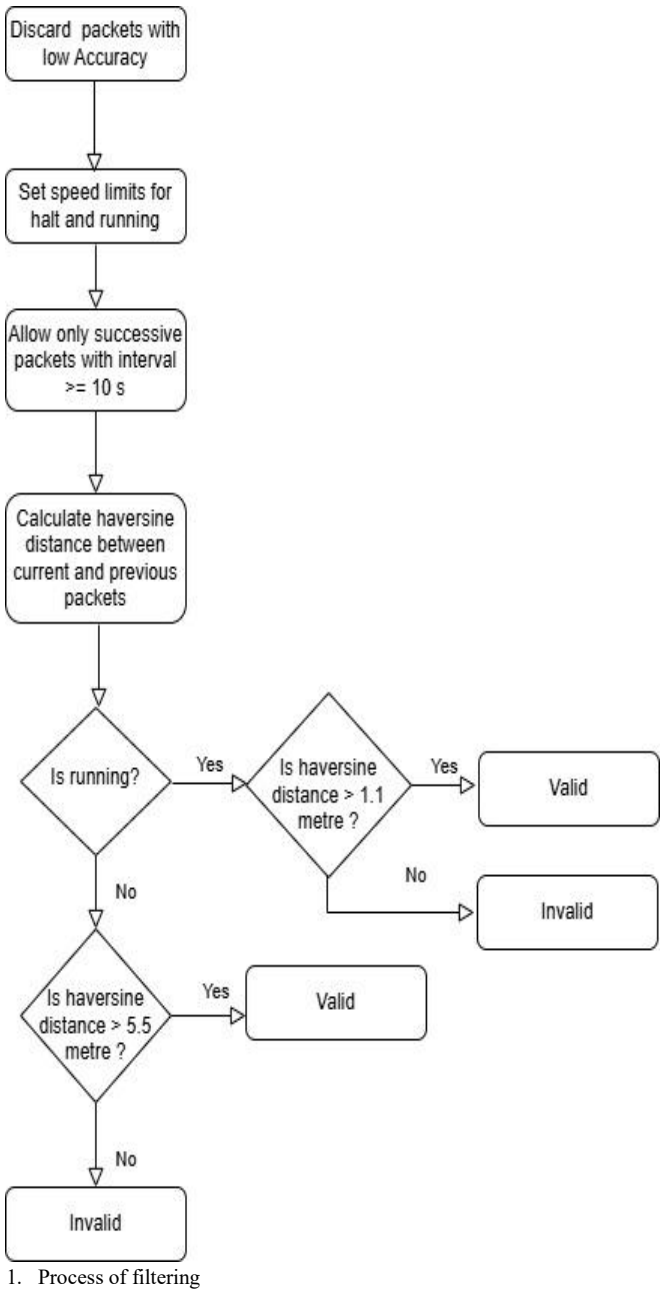



Fig. 1. Process of filtering



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### A. *Data cleaning*

The GPS accuracy varies with different environmental conditions. Signal blockage because of building, trees, indoor, under tunnels, bridges, and interference affects accuracy. The accuracy also depends upon the quality of device. The less accurate locations are not considered for plotting.

When the device is giving multiple packets within 10 second interval time, they are probable that those packets contribute to roaming or GPS wandering. So the packets beyond 10 seconds only are considered.

### B. *Setting speed limit*

The limit for speed should be different for halt and running vehicle. For running vehicle the speed is set beyond 5 Kilometre per hour and below the level it is considered as a halt. Vehicle in halt or running in low speed contributes more to GPS drift.

A running or a halt is decided by the above limit and the status of vehicle is set accordingly.

### C. *Filtering*

The data which has undergone cleaning is now filtered by calculating haversine distance between two successive packet location data with respect to UTC.

The threshold for halt is taken as 5.5 metres and for running it is taken as 1.1 metres. The locations less than these levels are marked as invalid and the others as valid. A column iscoordinatesvalid in database is updated as true when valid otherwise set as false. Only the valid locations of the vehicles are plotted on the map. So location coordinates appear in map if and only if the iscoordinatesvalid is true. Else it is not plotted.

The standard deviation of GPS receivers is in the range of one to five metres under open sky. The fluctuations or inaccuracy will be more while stationary or at lower speeds. Also while using latitude and longitude coordinates .00001 degrees is approximately equivalent to 1.11 metres. Based on these facts the above threshold is fixed.

## IV. Results And Discussion

The result shows location coordinates before and after filtration , sample results and plotted results.

### A. *Before filtration*

All the location coordinates that are available are plotted and shown here for analysis. The map showing all coordinates is in Fig. 2.

Numerous drift locations are displayed when on halt and also, less drift on running vehicle is shown in the map before filtration.

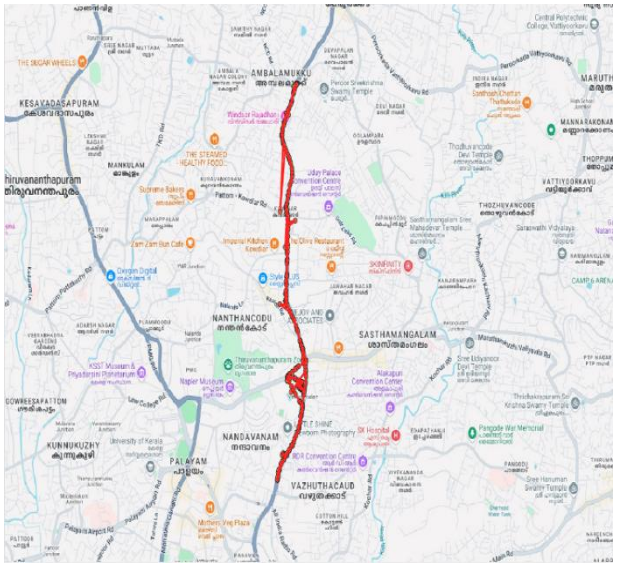



Fig. 2. Locations of vehicle in Google Map before Filtration

Fig. 3. map shows the focus on the areas of running and halt GPS drifts. The encircled with green shows drift on halt and with blue shows drift in running vehicle. The drift count for running as in this map is few whereas in halt most drift is noticed.

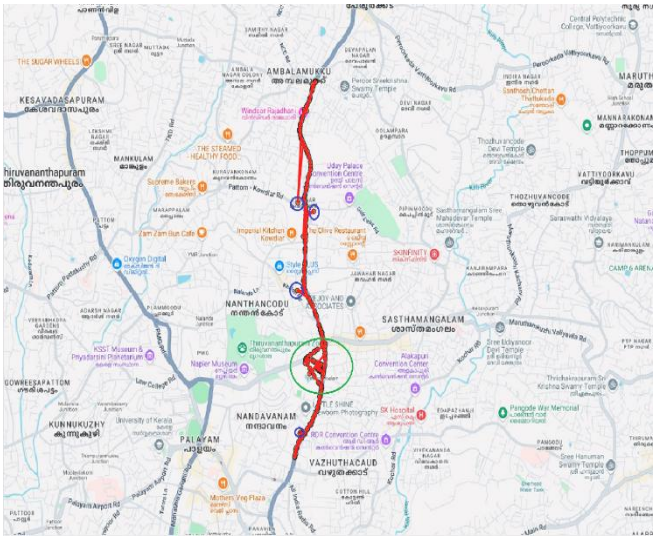



Fig. 3. Locations of vehicle in Google Map before Filtration showing drift encircled

### B. *After Filtration*

The results after filtration are as in Fig. 4. The location coordinates plotted here are of Smart phone-green dots, GPS-blue dots and MDT-red dots.



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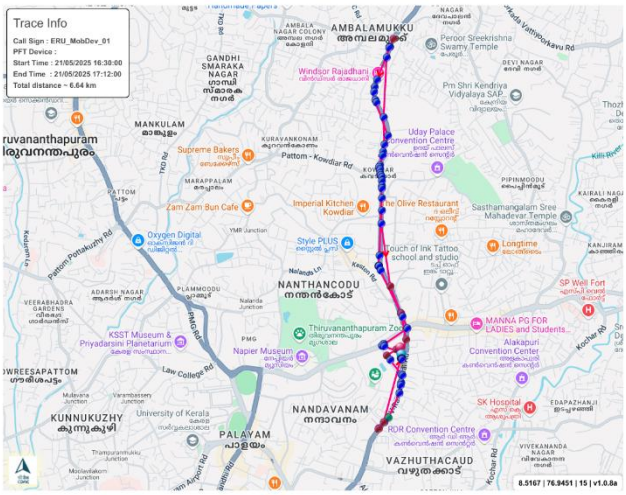



Fig. 4. After Filtration locations of vehicle plotted in Google Map screenshot from application

### C. *Result Analysis and Comparison*

*a) Distance and Validity:* The sample of data with location coordinates latitude, longitude, vehicle speed in Kilometres per hour (kmph), Haversine Distance in Metres (m) and validity set in iscoordinatesvalid column are presented in the following Table I.

TABLE I. SAMPLE RESULTS SHOWING DISTANCE AND VALIDITY

| **Latitude** | ***Longitude*** | ***Speed in kmph*** | ***Distance in m*** | ***iscoordinatesvalid*** |
|---|---|---|---|---|
| 8.5125661 | 76.9621348 | 5.73 | 344.661 | true |
| 8.509467 | 76.962192 | 0.2 | 0.566 | false |
| 8.509462 | 76.962193 | 0.2 | 0.646 | false |

*b) Unfiltered and filtered Locations:* The filtered and unfiltered locations data are shown with respect to count and percentage with details as given in the following Table II. It shows that the total number of locations is 127. After filtering out the 69 locations which are invalid and the resulting locations are 58 which are valid for plotting . For all invalid locations the iscordinatesvalid column is set to true and for invalid false. In terms of percentages, 46 percent locations which are invalid are left with and remaining 54 percent locations are filtered out.

TABLE II. DETAILS OF LOCATION COORDINATES AFTER FILTRATION

| **Location Coordinates** | ***Total*** | ***Unfiltered*** | ***Filtered Out*** |
|---|---|---|---|
| Count of values | 127 | 58 | 69 |
| Percentages | 100 | 46 | 54 |

Fig. 5. displays the results after filtration giving the count of filtered out and unfiltered locations.

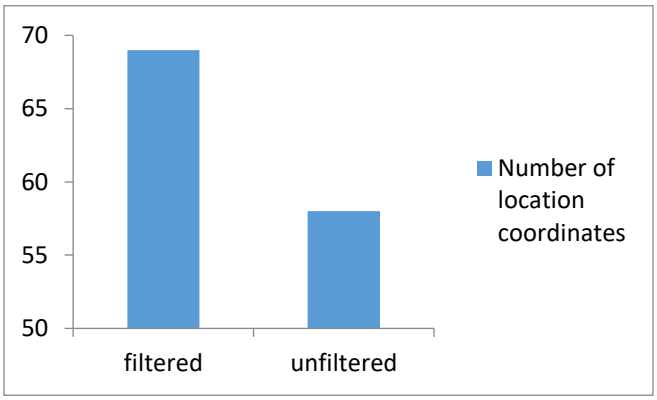



Fig. 5. Bar chart showing count of filtered out and unfiltered coordinates

Fig. 6. shows the percentages of filtered out and unfiltered location coordinates after filtration.

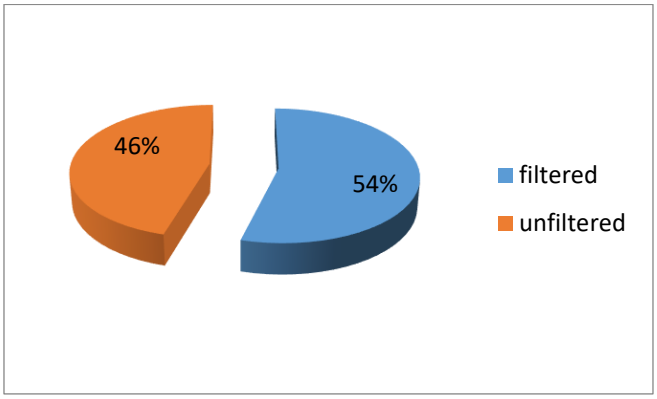



Fig. 6. Pie chart showing percentages of filtered out and unfiltered coordinates

From Fig. 4. it is visualized that GPS drift is reduced while in running and halt. As of with the 127 locations the 4 drifts while running are eliminated fully. The Drift while halting is not eliminated fully but it is minimized to a great extent.

## V. CONCLUSION

The appearance of GPS drift locations is eliminated to a larger level by employing the method stated in this paper. By not considering less accurate location coordinates, fixing speeds for halt and as well as a running vehicle, based on distance between successive location coordinates the GPS drift can be avoided to the most possible level. Thus it gives clear vehicle movement, better selection and monitoring, thereby reducing response time and hence better performance.

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